

Article

A Physics-Enhanced CNN–LSTM Predictive Condition Monitoring Method for Underground Power Cable Infrastructure

Zaki Moutassem ^{1,2}, Doha Bounaim ¹  and Gang Li ^{1,*} ¹ Michael W. Hall School of Mechanical Engineering, Mississippi State University, Starkville, MS 39762, USA; zm336@msstate.edu (Z.M.); db3068@msstate.edu (D.B.)² Underground Systems Inc., Bethel, CT 06801, USA

* Correspondence: gli@me.msstate.edu

Abstract

Underground high-voltage transmission cables, especially high-pressure fluid-filled (HPFF) pipe-type cable systems, are critical components of urban power networks. These systems consist of insulated conductor cables housed within steel pipes filled with pressurized fluids that provide essential insulation and cooling. Despite their reliability, HPFF cables experience faults caused by insulation degradation, thermal expansion, and environmental stressors, which, due to their subtle and gradual nature, complicate incipient fault detection and subsequent fault localization. This study presents a novel, proactive, and retrofit-friendly predictive condition monitoring method. It leverages distributed accelerometer sensors non-intrusively mounted on the HPFF steel pipe within existing manholes to continuously monitor vibration signals in real time. A physics-enhanced convolutional neural network–long short-term memory (CNN–LSTM) deep learning architecture analyzes these signals to detect incipient faults before they evolve into critical failures. The CNN–LSTM model captures temporal dependencies in acoustic data streams, applying time-series analysis techniques tailored for the predictive condition monitoring of HPFF cables. Experimental validation uses vibration data from a scaled-down HPFF laboratory test setup, comparing normal operation to incipient fault events. The model reliably identifies subtle changes in sequential acoustic patterns indicative of incipient faults. Laboratory experimental results demonstrate a high accuracy of the physics-enhanced CNN–LSTM architecture for incipient fault detection with effective data feature extraction. This approach aims to support enhanced operational resilience and faster response times without intrusive infrastructure modifications, facilitating early intervention to mitigate service disruptions.



Academic Editor: Ming-Feng Ge

Received: 17 August 2025

Revised: 23 September 2025

Accepted: 23 September 2025

Published: 25 September 2025

Citation: Moutassem, Z.; Bounaim, D.; Li, G. A Physics-Enhanced CNN–LSTM Predictive Condition Monitoring Method for Underground Power Cable Infrastructure. *Algorithms* **2025**, *18*, 600.

<https://doi.org/10.3390/a18100600>

Copyright: © 2025 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (<https://creativecommons.org/licenses/by/4.0/>).

Keywords: underground cables; predictive condition monitoring; fault detection; acoustic sensing; CNN–LSTM

1. Introduction

Underground transmission systems are a vital component of modern power infrastructure, particularly in urban areas where overhead lines face space and safety constraints. These systems offer advantages such as easier right-of-way acquisition and enhanced resilience to weather and vandalism. In addition, they help to preserve the visual character of urban areas, which often leads to strong public and municipal support in dense city environments [1–3]. To support these benefits, various types of underground power cables have been deployed, including cross-linked polyethylene, self-contained fluid-filled, and high-pressure fluid-filled (HPFF) or gas-filled (HPGF) cables. Each type of cable is

specifically designed to meet performance and reliability requirements [4–6]. In the United States, pipe-type cables comprise a significant portion of the underground high-voltage transmission network, many of which have now exceeded their 40-year design life, increasing the risk of faults [7,8]. Ensuring the reliability of these power cables is challenging, particularly in detecting and locating incipient faults (early-stage faults) before they cause service disruptions.

In contrast to overhead lines, where faults can often be identified visually or audibly [9], underground faults, whether open-circuit, short-circuit, or earth faults, remain concealed beneath the surface and are inherently difficult to detect and locate. These faults may arise from insulation breakdown caused by thermal and mechanical stresses along extended cable runs. Approximately 69% of faults, however, originate in cable accessories, most commonly due to installation errors [10–12]. Detecting underground cable faults, whether in accessories or along the cable run, is a critical first step, as this confirms their occurrence before localization methods are applied. Traditional fault detection approaches often rely on protection relay operations, circuit breaker trips, sudden current surges, or offline testing. However, these methods generally detect faults only after they become severe and often fail to capture incipient or intermittent failures. Once a fault is detected, locating it typically involves two steps: prelocation and pinpointing. Prelocation methods, such as time-domain reflectometry, arc reflection, and impedance-based techniques, estimate the fault distance from the terminals, but their accuracy diminishes over long cable runs [13–16]. Pinpointing then determines the exact location, traditionally by “thumping” the cable with high-voltage impulses and detecting acoustic signals generated at the fault location using surface-mounted geophones or microphones [17]. Although effective, thumping is labor-intensive and time-consuming, and may cause additional damage to the cable.

As demand grows for more resilient and intelligent underground transmission infrastructure, condition monitoring has emerged as a promising solution for incipient fault detection, localization, and diagnosis [18,19]. Existing approaches, including high-frequency signal analysis, traveling wave methods, and machine learning-based classifiers, have demonstrated effectiveness in improving incipient fault detection [20–24]. However, many of these methods rely on specialized instrumentation or offline testing, which limits their applicability for continuous, real-time monitoring. Recent developments in real-time distributed sensing for condition monitoring, such as fiber-optic methods like distributed temperature sensing (DTS) and distributed acoustic sensing (DAS), provide high spatial resolution but generally require pre-installed fibers, limiting their retrofit applicability. These limitations highlight the need for practical, scalable, retrofit-friendly monitoring solutions capable of detecting and localizing incipient faults in existing underground cables, beyond what traditional approaches can achieve.

Some deep learning-based time-series analysis methods, e.g., physics-informed neural networks (PINNs) and recurrent neural networks (RNNs), have been shown to be effective tools for fault detection and condition monitoring across various infrastructure systems [25]. PINNs incorporate the physical laws of complex systems into the neural network training process [26]. Although these physical laws are embedded in data-driven models, the standard architecture of PINNs limits their ability to effectively capture the evolving trends of dynamic processes [27]. Recent studies indicate that combining PINNs with data-driven models offers promising improvements in time-series prediction performance [28]. Long short-term memory (LSTM) networks, a type of RNN model, have been shown to process raw time-series operational data for accurate condition monitoring [29] and fault prognostics [30], highlighting their effectiveness in modeling temporal dependencies. LSTMs can also be incorporated with digital twins to predict incipient faults under extreme operational conditions [31]. Another type of RNN model is the Bidirectional LSTM (Bi-LSTM), which

uses both forward and backward sequences for enhanced pattern recognition [32]. Additionally, when combined with LSTM and attention mechanisms, a convolutional neural network (CNN), which extracts important features, enhances time-series classification accuracy and fault detection performance [33].

Our previous work [34] introduced a post-fault acoustic-based pinpointing method for HPFF cable systems, offering a promising alternative to conventional techniques. In this approach, accelerometers are temporarily mounted directly on the HPFF steel pipe in manholes near the pre-located fault zone. When a thumper is applied, repeated electrical discharges at the fault site generate stress waves that couple into the steel pipe wall, propagate along the pipe, and are captured by the sensors. By analyzing the time difference of arrival (TDOA) of these wavelets, the method aims to achieve high-accuracy fault localization in HPFF cable systems. This was demonstrated in a scaled-down laboratory setup, achieving centimeter-level pinpointing accuracy. While these results highlight the approach's potential to significantly improve pinpointing accuracy over traditional techniques, it remained inherently reactive, as it could only be deployed after a fault occurred and service disruption took place. Extending this approach, the present study transforms the concept into a permanently installed, non-intrusive distributed sensing system. Accelerometers are mounted directly onto the HPFF steel pipe within manholes along the transmission corridor, enabling continuous, real-time vibration monitoring. This upgraded framework is designed to support immediate detection and localization of full faults, while also providing the potential to detect incipient faults in real time without service disruption. By combining the precision of acoustic sensing with permanent installation, the method bridges the gap between post-fault pinpointing and proactive condition monitoring in underground power cables. Unlike fiber-optic systems, which often require continuous sensing media and infrastructure modifications, this newly developed sensing framework leverages existing manholes to retrofit accelerometers directly onto the steel pipe, eliminating the need for excavation or service disruptions during installation. As a result, it may enable scalable deployment with minimal impact on existing infrastructure to improve overall reliability and safety.

Building on this sensing framework, this study aims to address the critical need for incipient fault detection in existing HPFF cable systems and proposes a proactive condition monitoring solution capable of detecting incipient fault events before they escalate into complete failures. Incipient faults in these systems pose unique detection challenges due to their weak and intermittent signal characteristics, rendering conventional monitoring methods inadequate for timely identification. This study introduces a novel time-series condition monitoring methodology that combines physics-enhanced preprocessing with a hybrid CNN–LSTM architecture. This methodology leverages physical properties of wave propagation in steel pipe infrastructure to further enhance the detection capabilities of deep learning models. The physics-enhanced preprocessing incorporates TDOA to guide the selection of temporal windows for CNN–LSTM model training, ensuring that the neural network receives temporally consistent data that accurately reflects wave dynamics of HPFF cable systems. The primary objective is the real-time detection of incipient faults as they develop, enabling maintenance teams to implement preventive measures before system failures occur. Beyond its current demonstrated capability to detect faults in a controlled setting, this monitoring framework provides a foundation for the future development of real-time localization techniques for incipient faults. These advancements, once validated under real-world conditions, would enable operators to identify incipient fault locations more accurately, facilitating targeted interventions and minimizing service disruptions. Ultimately, this framework can extend the operational life of critical power

infrastructure, support more efficient maintenance planning, reduce operational costs for utilities, and provide a foundation for smarter, proactive grid management.

The remainder of this paper is organized as follows. Section 2 presents the physics-enhanced CNN–LSTM model, detailing the preprocessing methodology for high-frequency, multi-sensor data and the hybrid deep learning architecture. Section 3 describes the experimental setup and data collection procedures using a scaled-down HPFF cable testing system. Section 4 presents the results and discusses the performance evaluation of different architectural configurations. Finally, Section 5 concludes the study with key findings and future research directions.

2. Physics-Enhanced CNN–LSTM-Based Classification Condition Monitoring Model

2.1. Preprocessing of High-Frequency Multi-Sensor Data

The integrity and effectiveness of a data-driven incipient fault detection model rely heavily on the quality and structure of the input data. In this study, high-frequency acoustic vibration signals were recorded from four distributed sensors, yielding one million samples per sensor across a 20 s observation window, resulting in a high-resolution multi-sensor time-series dataset, as shown in Table 1. Prior to model training, raw data underwent statistical exploration, during which sensor #1 exhibited a high standard deviation of 5.97 and extreme outliers, with peak amplitudes reaching ± 700 mV, indicating its sensitivity to the simulated incipient fault event impacts, as shown in Figure 1. To address class imbalance in the high-frequency vibration dataset, an overlapping sliding window extraction method was implemented. The full temporal resolution of the dataset was reduced to 99,981 samples; this approach preserved all identifiable incipient fault segments while maximizing normal data diversity. The dataset was split into three categories: 70% training, 15% validation, and 15% testing. This controlled class imbalance and promoted robust generalization during model training by providing incipient fault samples in sufficient quantity and variation. The preprocessing stage was essential for enabling physics-enhanced deep learning architectures to effectively discriminate low-occurrence incipient fault events from normal system behavior in HPFF cable monitoring.

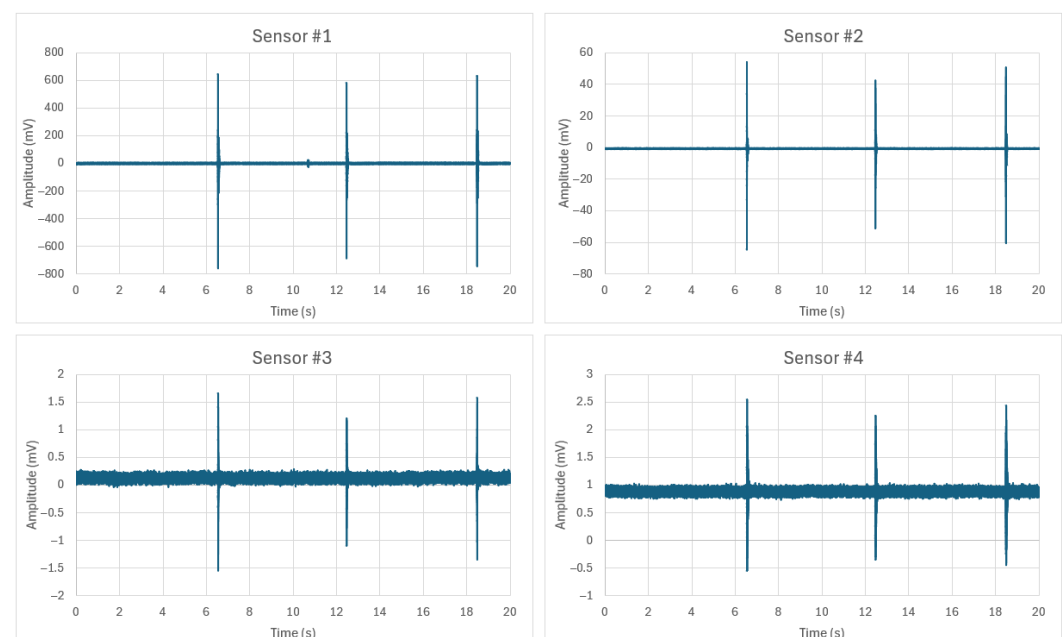


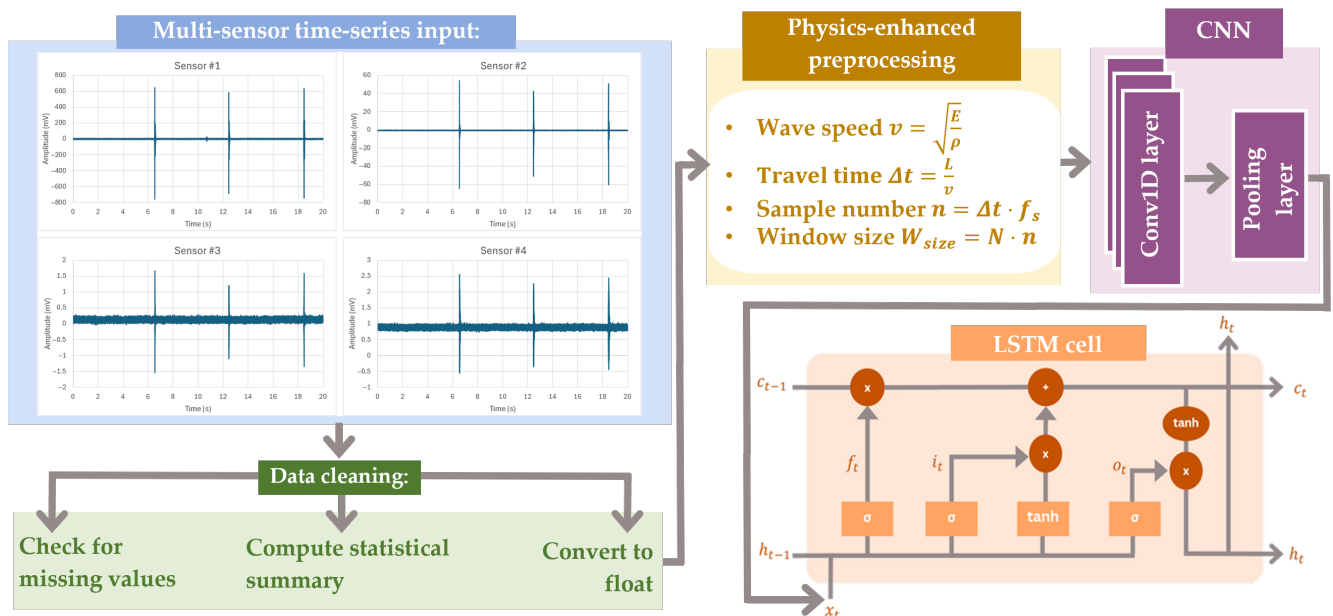
Figure 1. Multi-sensor vibration voltage measurements (mV) over a 20 s interval, showing recurrent incipient faults around 6.5 s, 12.5 s, and 18.5 s.

Table 1. Raw multi-sensor data for incipient fault detection in the HPFF cable system.

Incipient Fault/Normal	Time (s)	Sensor #1 (mV)	Sensor #2 (mV)	Sensor #3 (mV)	Sensor #4 (mV)
Normal events	0.00000	−1.03	−0.62	0.08	0.89
	0.00002	−1.01	−0.62	0.08	0.89
	0.00004	−1	−0.63	0.08	0.89
	0.00006	−0.98	−0.63	0.08	0.89
Incipient fault events	6.53524	153.82	3.66	0.48	0.93
	6.53526	172.99	−4.68	0.58	0.93
	6.53528	187.23	−13.78	0.64	0.92
	6.53530	195.67	−22.37	0.64	0.91
Normal events	19.99994	0.12	−0.75	0.07	0.84
	19.99996	0.13	−0.75	0.07	0.85
	19.99998	0.13	−0.73	0.07	0.86
	20.00000	0.15	−0.72	0.06	0.87

2.2. Physics-Enhanced Preprocessing

Figure 2 illustrates the proposed hybrid physics-enhanced CNN–LSTM framework that combines physics-enhanced preprocessing using TDOA with a CNN–LSTM architecture to enable the predictive condition monitoring of HPFF cable systems.

**Figure 2.** Physics-enhanced CNN–LSTM architecture for multi-sensor vibration-based incipient fault detection in underground HPFF cable systems.

The dataset comprises high-frequency vibration measurements collected from four sensors positioned at equal intervals along a 19 m long HPFF cable segment; details of the experimental setup and instrumentation are provided in Section 3. The signals were acquired at a high sampling frequency of 50 kHz to ensure the accurate temporal capture of transient waveforms generated by incipient fault events. Incipient faults were manually labeled on a voltage deviation threshold of ± 29.96 mV, and the incipient fault intervals were annotated to serve as reference labels for supervised learning. To integrate domain-specific knowledge, a physics-enhanced preprocessing approach was developed to determine

TDOA and define the temporal window size used in training. When an incipient fault occurs, waves propagate along the steel pipe with a velocity given as

$$v = \sqrt{\frac{E}{\rho}} \quad (1)$$

where E is the Young's modulus of steel and ρ is its density. The wave travel time between adjacent sensors is given as

$$\Delta t = \frac{L}{v} \quad (2)$$

where L is the length of the cable. The number of samples n to capture the time delay is given as

$$n = \Delta t \cdot f_s \quad (3)$$

where f_s is the sampling frequency. A physics-enhanced window is integrated to fully capture wave propagation across all three sensor intervals, which is

$$W_{\text{size}} = N \cdot n \quad (4)$$

where W_{size} is the window size and N is the number of sensor intervals. As a result, the model receives physically consistent input samples that mirror the real wave dynamics of HPFF cable systems. This strategy not only preserves incipient fault patterns but also reduces the risk of learning from truncated or incomplete signals, thereby improving model reliability and interpretability. A step size of 10 samples was used to extract overlapping windows to maximize data usage. The raw vibration data was segmented into overlapping windows of shape 188×4 , where 188 is the window size and 4 is the number of sensors. By integrating CNN, LSTM, and physics-enhanced preprocessing, the proposed method offers a practical and innovative solution for scenarios with limited data. The CNN is employed to extract key data features from high-frequency data, enabling the model to learn effectively even when training data is limited. Combined with LSTM and physics-enhanced windowing, this capability enables the reliable classification of subtle incipient faults while making efficient use of the minimal number of sensors and data. This design reflects real-world conditions, where sensors are sparsely deployed at manhole locations and fault events are inherently rare, demonstrating that the proposed architecture can capture the essential data features necessary for incipient fault detection.

2.3. CNN-LSTM Architectures

To perform the binary classification of incipient fault events, a hybrid deep learning model composed of CNN and LSTM networks was developed. In this architecture, CNN layers extract important information from the raw vibration signals, while LSTM layers model sequential dependencies and incipient fault propagation with the 188 sample windows derived from the physics-enhanced preprocessing.

2.3.1. LSTM

The LSTM is a type of recurrent neural network that addresses vanishing gradient issues by using gated memory units. At each time step t , the LSTM cell includes forget gate:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \quad (5)$$

input gate:

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \quad (6)$$

candidate memory:

$$g_t = \tanh(W_g \cdot [h_{t-1}, x_t] + b_g) \quad (7)$$

updated cell state:

$$c_t = f_t \odot c_{t-1} + i_t \odot g_t \quad (8)$$

output gate:

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \quad (9)$$

output hidden state:

$$h_t = o_t \odot \tanh(c_t) \quad (10)$$

where x_t is the raw vibration data input at time t , h_{t-1} is a previous hidden state, W_f , W_i , W_g , and W_o are the weight matrices for the forget gate, input gate, candidate memory, and output gate, respectively, b_f , b_i , b_g , and b_o are biases for the forget gate, input gate, candidate memory, and output gate, respectively, and σ is a sigmoid activation function.

2.3.2. CNN

CNNs serve as the feature extraction component in the hybrid CNN–LSTM architecture to identify local features within the multi-sensor vibration time-series data. CNN layers operate on the 188×4 input windows derived from the physics-enhanced preprocessing. They employ multiple convolutional layers with filters that slide across the temporal dimensions to detect characteristic incipient fault signals. The CNN model processes these windows using a one-dimensional CNN layer (Conv1D) at output channel j and time step t , which is given as

$$y_j(t) = \sum_{i=1}^{C_{in}} \sum_{k=0}^{K-1} W_{k,i,j} \cdot x_i(t-k) + b_j \quad (11)$$

where $W_{k,i,j}$ is the weight of the filter at position k for input channel i and output channel j , K is the kernel size, and b_j is the bias term for j .

3. Experimental Setup and Data Processing

3.1. Experimental Setup and Data Collection

The experimental setup in this study is identical to the fully embedded configuration in our previous work [34], representing a scaled-down model of an actual HPFF pipe-type cable system. For clarity and completeness, the schematic of the scaled-down experimental setup in Figure 3 describes the layout of sensors for the tested HPFF pipe-type cable system. A 19 m long Schedule-80 carbon steel pipe (inner diameter 49.25 mm), coated externally with Pritec (HDPE over butyl mastic), was placed centrally inside a containment box measuring $17 \text{ m} \times 0.46 \text{ m} \times 0.61 \text{ m}$. The box was lined with vinyl and aluminum foil at the base for grounding purposes. The steel pipe was embedded within compacted rock dust, selected for its thermal conductivity and compaction properties typical of HPFF backfill. The rock dust was added and tamped in 5–8 cm lifts until the pipe was fully surrounded, with access manholes provided for sensor placement at intermediate locations, as shown in Figure 4.

Four Brüel & Kjær (B&K) 4518-002 accelerometers were attached to the pipe at equally spaced intervals ($\sim 6.33 \text{ m}$) using beeswax. The accelerometers were connected to B&K 1704-A-002 signal conditioners and a Delphin Expert Transient Data Logger (detailed specifications in Tables 2–5). Sensor calibration and time synchronization were confirmed through co-located testing with reference acoustic pulses, as described in [34].

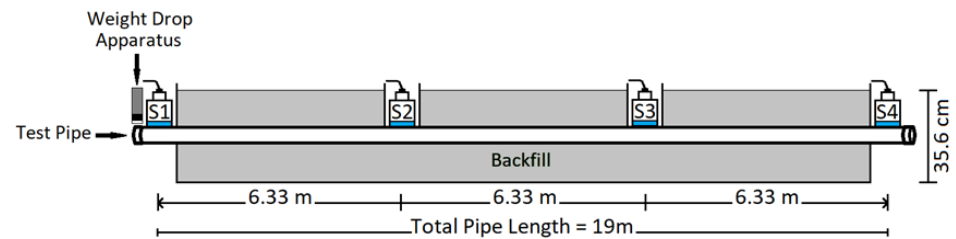


Figure 3. Schematic of the experimental setup of the scaled-down HPFF pipe-type cable. Incipient faults were simulated at the Sensor 1 location using a weight drop apparatus.

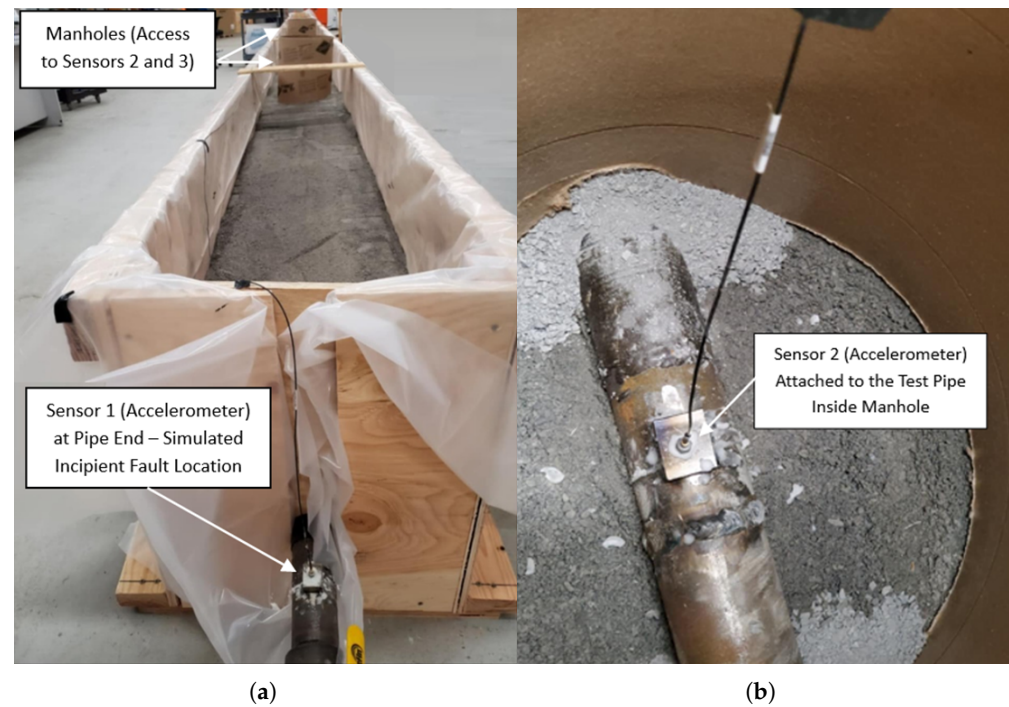


Figure 4. Experimental setup of the scaled-down HPFF pipe-type cable system: (a) overview of full setup and (b) manhole access points at intermediate locations for accelerometer placement.

Table 2. Detailed parameters of the HPFF pipe-type cable system.

Item	Value
Pipe length (d)	19 m
Material	Carbon steel
Young's modulus (E)	200 GPa
Density (ρ)	7850 kg/m ³
Outer diameter (OD)	60.33 mm
Inner diameter (ID)	49.25 mm
Wall thickness	5.54 mm (Sch. 80)
External coating	Pritec
Internal coating	Epoxy

Acoustic pulses were generated using a weight drop apparatus that released a 0.91 kg weight from a specified height onto the pipe at Location 1 (one pipe end). Five repeated runs were conducted, capturing both baseline and anomaly (incipient fault) pulse data. A 3 kHz low-pass filter was applied to recorded signals to reduce noise while preserving pulse characteristics. This dataset was used to train the physics-enhanced CNN–LSTM model for predictive condition monitoring to detect incipient faults within the embedded pipe system. The primary goal was to establish a proof-of-concept for the physics-enhanced CNN–LSTM approach for effective data feature extraction and incipient fault detection under controlled, scaled-down HPFF laboratory testing conditions.

Table 3. Detailed parameters of B&K accelerometers.

Item	Value
Product model	B&K 4518-002
Sensitivity	$10 \pm 10\%$ mV/g *
Measurement range	± 500 g
Resonant frequency	62 kHz
Frequency range	1–20,000 Hz
Residual noise level	2000 μ g
Maximum operational level (peak)	500 g
Mounting	Adhesive

* g is the acceleration of gravity.

Table 4. Detailed parameters of the B&K CCLD signal conditioner.

Item	Value
Product model	B&K 1704-A-002
Maximum frequency	55 kHz
Minimum frequency	2.2 Hz
Maximum gain (dB)	$\times 100$ (40 dB)
Minimum gain (dB)	$\times 1$ (0 dB)

Table 5. Specifications of the Delphin Expert Transient Data Logger.

Item	Value
Product model	Delphin Expert Transient
Number of input channels	4
Number of output channels	8
Voltage range	± 25 V
Measurement accuracy	0.5 mV + 0.008%
Max. input frequency/min. pulse width	1 MHz/500 ns
Sampling rate	20–50 kHz

3.2. Experimental Data Processing

The collected acoustic data encompassed both normal operational conditions and simulated incipient fault scenarios. Normal conditions represented baseline vibration measurements during steady-state operation, while incipient fault conditions were induced through controlled impact events at specific locations along the pipe system. This controlled approach ensured consistent and repeatable incipient fault signatures for model training and validation.

The high-frequency sampling rate of 50 kHz captured transient phenomena with sufficient temporal resolution to distinguish between background vibrations observed in the laboratory environment, e.g., vibrations from the lab structure or equipment operation, and incipient fault-related acoustic emissions. Signal preprocessing includes a 3 kHz low-pass filter to remove most high-frequency noise while preserving the characteristic frequency content of incipient fault-related events.

3.3. Hyperparameter Settings for Physics-Enhanced CNN–LSTM

Hyperparameters of the physics-enhanced CNN–LSTM model were configured to optimize incipient fault detection performance while maintaining computational efficiency. Nine different architectural configurations were evaluated in Section 4 by varying the number of CNN and LSTM layers to determine the optimal balance between complexity and classification accuracy. The physics-enhanced preprocessing achieved a window size of 188 samples to maximize data utilization and preserve temporal dependencies. The training employed early stopping with a patience of 3 epochs to prevent overfitting. A systematic grid search across the nine architectural configurations was employed for fine-tuning the CNN–LSTM hyperparameters within the optimal performance, as detailed in Table 6.

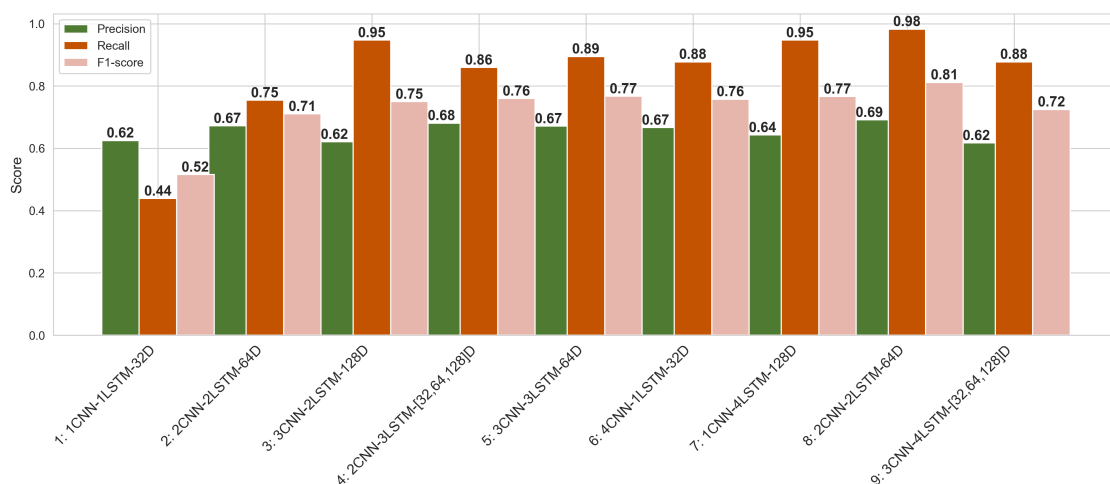
Table 6. Optimized hyperparameters of the physics-enhanced CNN–LSTM model.

	Parameter	Value
Model architecture	CNN layer	3
	CNN filter	32
	MaxPooling size	2
	LSTM layers	2
	LSTM neurons	64
	Dense layer	128
Training parameters	Batch size	64
	Epoch	30
	Early stopping patience	3 epochs

4. Results and Discussion

4.1. Evaluation Metrics and Results

Due to high class imbalance, where normal signals vastly outnumber incipient fault signals, reflecting real-world conditions, model accuracy alone was not considered a sufficient performance metric. Instead, additional metrics, i.e., F1-score, recall incipient faults, precision incipient faults, false positive rate (FPR), false negative rate (FNR), and training time, were taken into consideration when selecting hyperparameters. This contrasts with previous studies that primarily focused on maximizing the overall accuracy of models. As shown in Figures 5–7, the third configuration of three CNN layers and two LSTM layers demonstrated the most effective balance between classification performance and computational efficiency, achieving strong performance metrics, with precision, recall, and F1-scores of 0.62, 0.95 and 0.75, respectively, while maintaining one of the shortest training times at only 240 s. This configuration showed promising results that justified its selection as an optimal architecture for employment. As the architectural complexity increased through deeper networks, the improvement in classification metrics was marginal and did not justify the dramatic increase in computational cost. For example, while configuration 7 achieved minimal performance gains, the training time increased to 3092 s, making the third configuration the most practical choice for real-time cable monitoring applications where both accuracy and computational efficiency are critical components.

**Figure 5.** Classification performance metrics (precision, recall, F1-score) across nine CNN–LSTM architectural configurations.

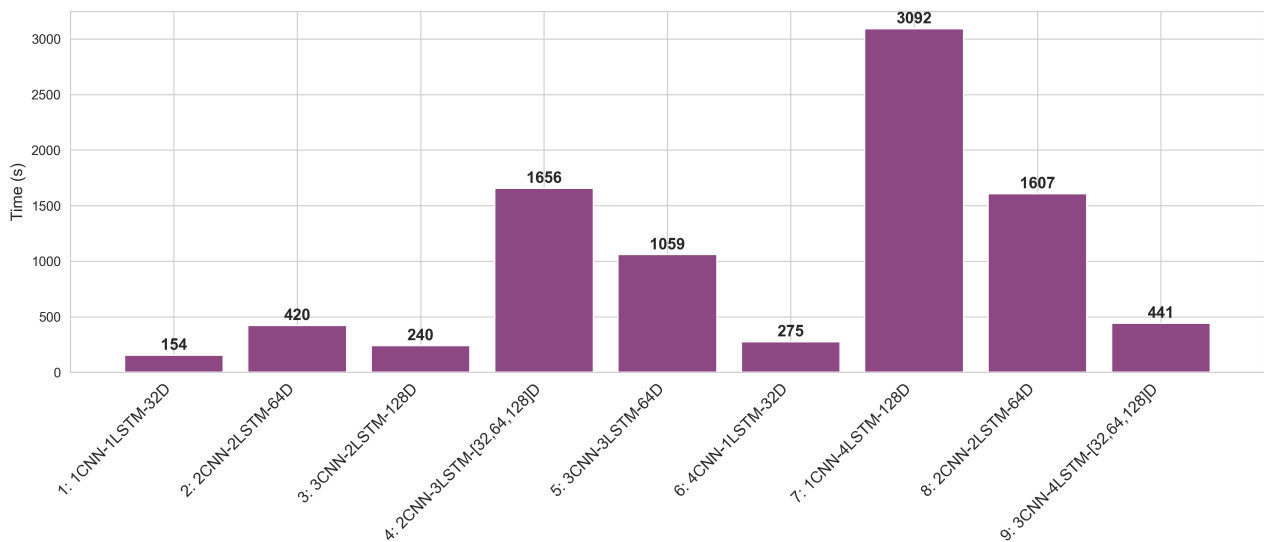


Figure 6. Training time comparison across nine CNN–LSTM architectural configurations.

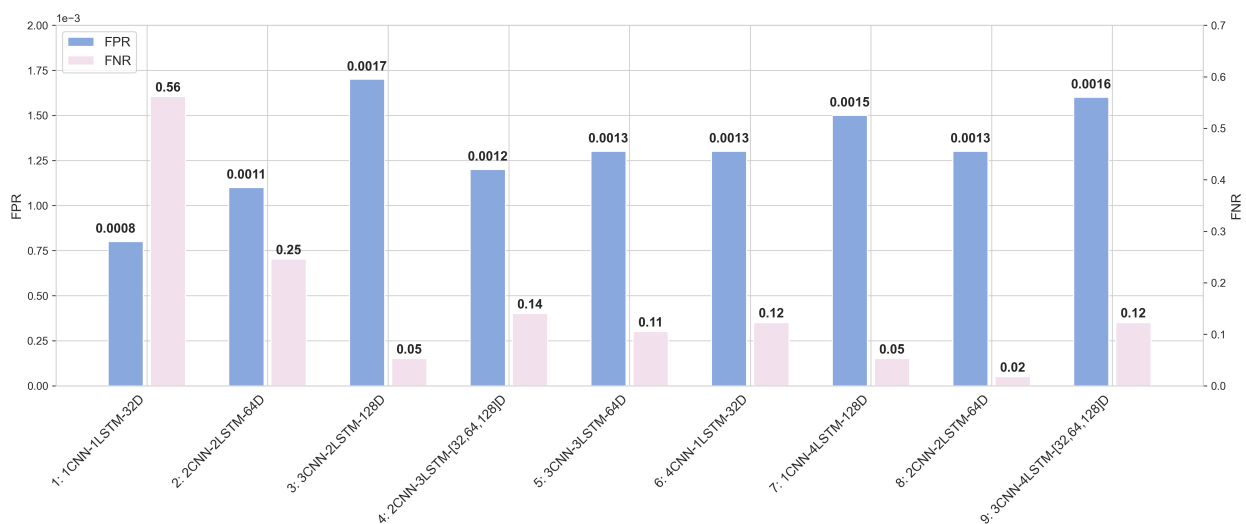


Figure 7. Error rate analysis (FPR and FNR) across nine CNN–LSTM architectural configurations.

4.2. Comparison with Other Time-Series Models

Table 7 presents a comprehensive comparison between the optimized CNN–LSTM model and traditional machine learning approaches for incipient fault detection in HPFF cable systems, each incorporating physics-enhanced preprocessing. The proposed physics-enhanced CNN–LSTM architecture vastly outperforms all traditional deep learning models across all evaluation metrics. The CNN–LSTM achieved an incipient fault precision of 0.6914, incipient fault recall of 0.9825, and incipient fault F1-score of 0.8116, demonstrating the accurate identification of incipient fault events while minimizing FPR. The CNN–LSTM model demonstrated over 59% improvement in F1-score over the LSTM model and a stronger improvement of 132% compared to the artificial neural network (ANN) model.

Figure 8 provides detailed insights into the classification behavior of different deep learning models through confusion matrix analysis. The CNN–LSTM model’s confusion matrix demonstrates superior incipient fault detection capabilities with only 17 false negatives, achieving a low FNR despite the dataset’s high proportion of normal conditions that closely simulate real-world scenarios where incipient fault events are rare occurrences. This performance in minimizing missed detections is critical for safe condition monitoring,

where undetected incipient faults can lead to service interruptions and pose significant safety risks.

Table 7. Comparison of standard models compared to CNN–LSTM.

Model	Precision (Incipient Fault)	Recall	F1-Score (Incipient Fault)
ANN	0.4700	0.2800	0.3500
RNN	0.5000	0.0700	0.1200
LSTM	0.6500	0.4200	0.5100
CNN–LSTM	0.6914	0.9825	0.8116

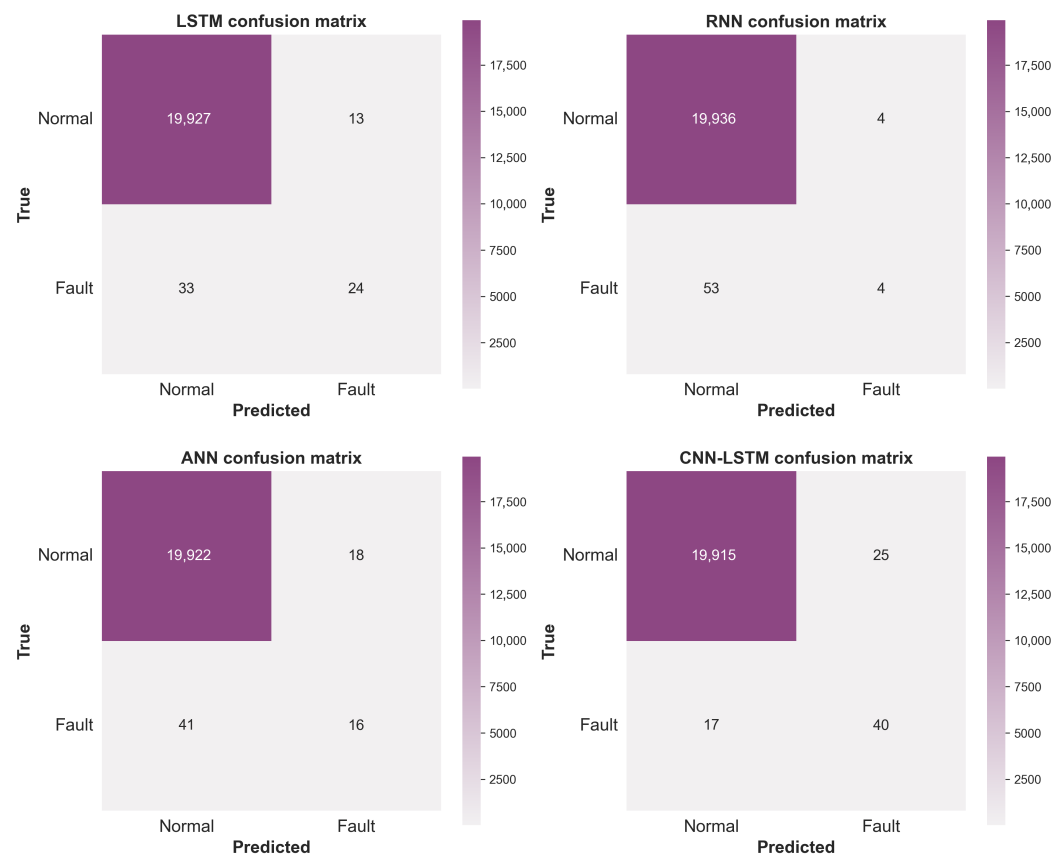


Figure 8. Confusion matrix comparison of deep learning models for incipient fault classification in HPFF cable systems.

5. Conclusions

This work presents a physics-enhanced CNN–LSTM framework for the predictive condition monitoring of HPFF cable systems, advancing beyond purely reactive post-fault detection to an integrated approach that supports continuous monitoring, incipient fault detection, and the potential for rapid, subsequent fault localization. By embedding wave propagation physics into the preprocessing stage, the proposed method generates temporally consistent input sequences that accurately capture vibration dynamics, enhancing interpretability and demonstrating promising reliability in detecting incipient faults.

Validation on a 19 m laboratory-scale HPFF cable system demonstrated that the proposed physics-enhanced CNN–LSTM architecture achieved an incipient fault detection F1-score of 0.8116. This substantially outperformed baseline models, including ANN (F1-score 0.35), RNN (0.12), and standalone LSTM (0.51). Notably, the proposed model attained an exceptionally high recall of 0.9825 in the experimental setup, indicating strong potential

for detecting true incipient fault events while maintaining computational efficiency suitable for real-time monitoring.

To further fully capture the complexity of real underground HPFF cable networks, future work will focus on several key directions: (i) conducting large-scale tests with longer cables, additional sensors, and realistic urban operational conditions, including broadband noise, to fully evaluate performance; (ii) implementing experiments on a real HPFF cable system to validate the framework's robustness and its ability to detect subtle incipient faults under practical operating conditions; and (iii) exploring advanced physics-informed neural network techniques to improve detection performance, enhance interpretability, and extend capabilities toward precise incipient fault localization, addressing potential improvements in the physics-enhanced CNN–LSTM model. These extensions will be explored as separate, comprehensive studies to preserve clarity and depth in each contribution.

The approach proposed in this study shows potential as a retrofit-friendly solution that leverages existing manhole access points to minimize installation disruption and provide incipient fault warning capabilities. Overall, the complete framework offers a promising pathway toward the smarter, more reliable condition monitoring of HPFF cable systems, with the potential to enhance grid resilience, reduce outages, accelerate diagnostics, and extend asset life.

Author Contributions: Conceptualization, Z.M. and G.L.; methodology, Z.M. and D.B.; software, Z.M.; validation, Z.M. and G.L.; formal analysis, Z.M. and D.B.; investigation, Z.M. and D.B.; resources, Z.M.; data curation, Z.M.; writing—original draft preparation, Z.M. and D.B.; writing—review and editing, G.L.; visualization, Z.M. and D.B.; supervision, G.L.; project administration, G.L.; funding acquisition, Z.M. and G.L. All authors have read and agreed to the published version of the manuscript.

Funding: This research was funded by the U.S. Department of Energy, grant number DE-SC0012070, and the U.S. National Science Foundation, grant numbers 2329791 and OIA-2429540.

Data Availability Statement: The original contributions presented in this study are included in the article. Further inquiries can be directed to the corresponding authors.

Acknowledgments: The authors are grateful for the materials and experimental testing support from Underground Systems Inc. (USi).

Conflicts of Interest: The first author was employed by the company of Underground Systems Inc. The remaining authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as potential conflicts of interest.

References

1. Bumby, S.; Druzhinina, E.; Feraldi, R.; Werthmann, D.; Geyer, R.; Sahl, J. Life cycle assessment of overhead and underground primary power distribution. *Environ. Sci. Technol.* **2010**, *44*, 5587–5593. [\[CrossRef\]](#)
2. Swain, A.; Abdellatif, E.; Mousa, A.; Pong, P.W.T. Sensor technologies for transmission and distribution systems: A review of the latest developments. *Energies* **2022**, *15*, 7339. [\[CrossRef\]](#)
3. Trakas, D.N.; Hatziaargyriou, N.D. Strengthening transmission system resilience against extreme weather events by undergrounding selected lines. *IEEE Trans. Power Syst.* **2022**, *37*, 2808–2820. [\[CrossRef\]](#)
4. Tziouvaras, D. Protection of high-voltage AC cables. In Proceedings of the 59th Annual Conference for Protective Relay Engineers, 2006, College Station, TX, USA, 4–6 April 2006; pp. 48–61. [\[CrossRef\]](#)
5. Ogbogu, O. Review on enhancement of electric power transmission using underground cabling. *Int. J. Res. Appl. Sci. Eng. Technol.* **2019**, *7*, 519–524. [\[CrossRef\]](#)
6. Ohno, T. Various cables used in practice. In *Cable System Transients: Theory, Modeling and Simulation*; John Wiley & Sons, Ltd.: Hoboken, NJ, USA, 2015; pp. 1–20.
7. Eckroad, S.; Institute, E.P.R. *EPRI Underground Transmission Systems Reference Book*; Electric Power Research Institute: Palo Alto, CA, USA, 2007.

8. Kleinfelder, H. Assessing the integrity and increasing the service life of HPFF pipe type cables. In Proceedings of the 2011 IEEE PES 12th International Conference on Transmission and Distribution Construction, Operation and Live-Line Maintenance (ESMO), Providence, RI, USA, 16–19 May 2011; pp. 1–4. [\[CrossRef\]](#)
9. Zhang, X.; Yang, J.; Zhu, W.; Li, G. A non-destructive health monitoring method for wooden utility poles with frequency-modulated empirical mode decomposition and Laplace wavelet correlation filtering. *Sensors* **2022**, *22*, 4007. [\[CrossRef\]](#)
10. Li, Z.; Zhou, K.; Wang, C.; Meng, P.; Li, Y.; Lin, S.; Fu, Y.; Yuan, H. Failure of cable accessory: Interface breakdown under the effect of moisture. *IEEE Trans. Power Deliv.* **2024**, *39*, 2644–2652. [\[CrossRef\]](#)
11. Shaalan, E.M.; Ward, S.A.; Youssef, A. Analysis of a practical study for underground cable faults causes. In Proceedings of the 2021 22nd International Middle East Power Systems Conference (MEPCON), Assiut, Egypt, 14–16 December 2021; pp. 208–215. [\[CrossRef\]](#)
12. Ali, K.H.; Bradley, S.; Aboushady, A.A.; Abdel Maksoud, S.A.; Farrag, M.E. Developing a framework for underground cable fault-finding in low voltage distribution networks. In Proceedings of the 2020 9th International Conference on Renewable Energy Research and Application (ICRERA), Glasgow, UK, 27–30 September 2020; pp. 477–482. [\[CrossRef\]](#)
13. Ali, K.H.; Aboushady, A.A.; Bradley, S.; Farrag, M.E.; Abdel Maksoud, S.A. An industry practice guide for underground cable fault-finding in the low voltage distribution network. *IEEE Access* **2022**, *10*, 69472–69489. [\[CrossRef\]](#)
14. Abd-Elaziz, A.A.; Khan, S.; Aboushady, A.A.; Farrag, M.E.; Merlin, M.M.C.; Finney, S.; Abdel Maksoud, S. Fault pinpointing in underground cables of low-voltage distribution networks with inductive wireless power transfer. *Energies* **2024**, *17*, 6304. [\[CrossRef\]](#)
15. Personal, E.; García, A.; Parejo, A.; Larios, D.F.; Biscarri, F.; León, C. A comparison of impedance-based fault location methods for power underground distribution systems. *Energies* **2016**, *9*, 1022. [\[CrossRef\]](#)
16. Gururajapathy, S.; Mokhlis, H.; Illias, H. Fault location and detection techniques in power distribution systems with distributed generation: A review. *Renew. Sustain. Energy Rev.* **2017**, *74*, 949–958. [\[CrossRef\]](#)
17. Lanz, B.; Sanchez, E. Is Fault Location Killing Our Cable Systems? In Proceedings of the 2016 IEEE/PES Transmission and Distribution Conference and Exposition (T&D), Dallas, TX, USA, 3–5 May 2016; pp. 1–5. [\[CrossRef\]](#)
18. Khan, A.; Malik, N.; Al-Arainy, A.; Alghuwainem, S. A review of condition monitoring of underground power cables. In Proceedings of the 2012 IEEE International Conference on Condition Monitoring and Diagnosis, Bali, Indonesia, 23–27 September 2012; pp. 909–912. [\[CrossRef\]](#)
19. Carmo, E.; da Silva, L.; Maia, T. Survey on incipient fault localization methods in underground cables. *Comput. Electr. Eng.* **2025**, *123*, 109961. [\[CrossRef\]](#)
20. Das, M.; Mishra, S.; Swain, S.; Biswal, T. A review on incipient fault detection, location and classification in underground cable. In Proceedings of the International Conference on Energy Systems, Drives and Automations, Kolkata, India, 31 December 2022; pp. 173–182. [\[CrossRef\]](#)
21. Li, Q.; Luo, H.; Cheng, H.; Deng, Y.; Sun, W.; Li, W.; Liu, Z. Incipient fault detection in power distribution system: A time-frequency embedded deep-learning-based approach. *IEEE Trans. Instrum. Meas.* **2023**, *72*, 2507914. [\[CrossRef\]](#)
22. Robson, S.; Haddad, A.; Griffiths, H. Traveling wave fault location using layer peeling. *Energies* **2019**, *12*, 126. [\[CrossRef\]](#)
23. Ghanbari, T. Kalman filter based incipient fault detection method for underground cables. *IET Gener. Transm. Distrib.* **2015**, *9*, 1988–1997. [\[CrossRef\]](#)
24. Sahoo, P.; Mishra, S.; Das, M.; Swain, S. Enhanced incipient fault identification in underground distribution cable implementing Random Forest classifier. *Microsyst. Technol.* **2025**, *31*, 1763–1773. [\[CrossRef\]](#)
25. Yang, B.; Liang, X.; Xu, S.; Wong, M.; Ma, W. A time-series based deep survival analysis model for failure prediction in urban infrastructure systems. *Eng. Appl. Artif. Intell.* **2024**, *136*, 108876. [\[CrossRef\]](#)
26. Farea, A.; Yli-Harja, O.; Emmert-Streib, F. Understanding physics-informed neural networks: Techniques, applications, trends, and challenges. *AI* **2024**, *5*, 1534–1557. [\[CrossRef\]](#)
27. Velioglu, M.; Zhai, S.; Rupprecht, S.; Mitsos, A.; Jupke, A.; Dahmen, M. Physics-informed neural networks for dynamic process operations with limited physical knowledge and data. *Comput. Chem. Eng.* **2025**, *192*, 108899. [\[CrossRef\]](#)
28. Wu, Y.; Sicard, B.; Gadsden, S. Physics-informed machine learning: A comprehensive review on applications in anomaly detection and condition monitoring. *Expert Syst. Appl.* **2024**, *255*, 124678. [\[CrossRef\]](#)
29. Barbre, Z.; Li, G. Enhanced wind energy forecasting using an extended long short-term memory model. *Algorithms* **2025**, *18*, 206. [\[CrossRef\]](#)
30. Afridi, Y.; Hasan, L.; Ullah, R.; Ahmad, Z.; Kim, J. LSTM-based condition monitoring and fault prognostics of rolling element bearings using raw vibrational data. *Machines* **2023**, *11*, 531. [\[CrossRef\]](#)
31. Liu, Y.; Shangquan, D.; Chen, L.; Liu, X.; Yin, G.; Li, G. Multi-domain digital twin and real-time performance optimization for marine steam turbines. *Symmetry* **2025**, *17*, 689. [\[CrossRef\]](#)

32. Malashin, I.; Tynchenko, V.; Gantimurov, A.; Nelyub, V.; Borodulin, A. Applications of long short-term memory (LSTM) networks in polymeric sciences: A review. *Polymers* **2024**, *16*, 2607. [[CrossRef](#)]
33. Borré, A.; Seman, L.; Camponogara, E.; Stefenon, S.; Mariani, V.; Coelho, L. Machine fault detection using a hybrid CNN-LSTM attention-based model. *Sensors* **2023**, *23*, 4512. [[CrossRef](#)] [[PubMed](#)]
34. Moutassem, Z.; Li, G.; Zhu, W. Experimental investigation of steel-borne acoustic pulses for fault pinpointing in pipe-type cable systems: A scaled-down model approach. *Sensors* **2024**, *24*, 7043. [[CrossRef](#)] [[PubMed](#)]

Disclaimer/Publisher’s Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.